

# torch.nan\_to\_num#

[https://pytorch.org/docs/stable/generated/torch.nan\\_to\\_num.html#torch.nan\\_to](https://pytorch.org/docs/stable/generated/torch.nan_to_num.html#torch.nan_to)

## Description:

Replaces NaN, positive infinity, and negative infinity values in input with the values specified by nan, posinf, and neginf, respectively. By default, NaNs are replaced with zero, positive infinity is replaced with the greatest finite value representable by input's dtype, and negative infinity is replaced with the least finite value representable by input's dtype. input (Tensor) – the input tensor. nan (Number, optional) – the value to replace NaNs with. Default is zero. posinf (Number, optional) – if a Number, the value to replace positive infinity values with. If None, positive infinity values are replaced with the greatest finite value representable by input's dtype. Default is None. neginf (Number, optional) – if a Number, the value to replace negative infinity values with. If None, negative infinity values are replaced with the lowest finite value representable by input's dtype. Default is None.

## Function Signature

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```
torch.nan_to_num(input, nan=0.0, posinf=None, neginf=None, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

## Parameters

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### Keyword

out (Tensor, optional) – the output tensor.

## Code Examples

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```
>>> x = torch.tensor([float('nan'), float('inf'), -float('inf'), 3.14])
>>> torch.nan_to_num(x)
tensor([ 0.0000e+00,  3.4028e+38, -3.4028e+38,  3.1400e+00])
>>> torch.nan_to_num(x, nan=2.0)
tensor([ 2.0000e+00,  3.4028e+38, -3.4028e+38,  3.1400e+00])
>>> torch.nan_to_num(x, nan=2.0, posinf=1.0)
tensor([ 2.0000e+00,  1.0000e+00, -3.4028e+38,  3.1400e+00])
```

# Detailed Documentation

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## Documentation

Replaces NaN, positive infinity, and negative infinity values in input with the values specified by nan, posinf, and neginf, respectively. By default, NaNs are replaced with zero, positive infinity is replaced with the greatest finite value representable by input's dtype, and negative infinity is replaced with the least finite value representable by input's dtype.

input (Tensor) – the input tensor.

nan (Number, optional) – the value to replace NaNs with. Default is zero.

posinf (Number, optional) – if a Number, the value to replace positive infinity values with. If None, positive infinity values are replaced with the greatest finite value representable by input's dtype. Default is None.

neginf (Number, optional) – if a Number, the value to replace negative infinity values with. If None, negative infinity values are replaced with the lowest finite value representable by input's dtype. Default is None.

out (Tensor, optional) – the output tensor.